

Social Context of Computing

Learning Objectives

- Interpret the social context of a particular software/hardware implementation
- Computing tool implementation
- Positive and negative ways in computing
- Why computing/network access is restricted in some countries
- Impact of the digital divide
- Understand how income, geography, race, and culture influence access to information technology and technology in general
- Identify how internet has changed the face of computing and it has affected society



Young and aggressive, **Johnny** has a promising future as a business. Johnny is on track to reach a million dollars before the age of 25 thanks to several successful businesses and a few start-ups with promising financial standing.

Johnny's business talks are basically in **difficult areas**. Because of this, Johnny employs a group of trained security bodyguards to follow him nearly 24 hours a day so that he can feel safe. A constant stream of visitors, including both friends and business partners, visit Johnny in his expansive \$10 million house. Along with protecting him, his bodyguards also ensure that visitors are welcomed and escorted out of the venue in an orderly manner.

As a result, the bodyguards maintain a constant office and sleeping accommodations at Johnny's mansion. Johnny equipped the guards' offices and sleeping accommodations with cameras and listening devices without telling them in order to record every conversation and movement. He is secure in that manner!



1. Is Johnny violating any law ?
2. Do the bodyguards have any right to privacy on Johnny ' s premises ?
3. Does Johnny have a right to know what the bodyguards are doing in their private quarters?



Introduction

- Our focus in this chapter is on both the social and ethical effects of computer technology on people, whether we are at home, school, or work. We will focus on the social and economic dimensions of computing as a result of the “digital divide,” the workplace, workplace monitoring of employees, and the well-being of employees.





Social Categories Impacted By Technology

Socio-demographic Groups

Affecting social/
economic class,
race, and gender

Social And
Political
Institutions

Such as education
and government

Social Sectors

Including the
workplace

<https://www.slideshare.net/UpekhaVandebona/social-context-of-computing>

Digital Divide

- The **digital divide** is a term that refers to the gap between demographics and regions that have access to modern information and communications technology (ICT), and those that don't or have restricted access. This technology can include the telephone, television, personal computers and internet connectivity.
- In general, the study of the digital divide involves the study of the impact of the digital divide indicators. These indicators concern communication technologies such as radio, television, the press, fixed and cellular telephones, fax machines, computers, and connectivity to the Internet and participation in cyber activities for all members of a society.



5 indicators of Digital Divide

- Access
- Relevant technology
- Human ware (human capacity)
- Infrastructure
- Enabling environment .



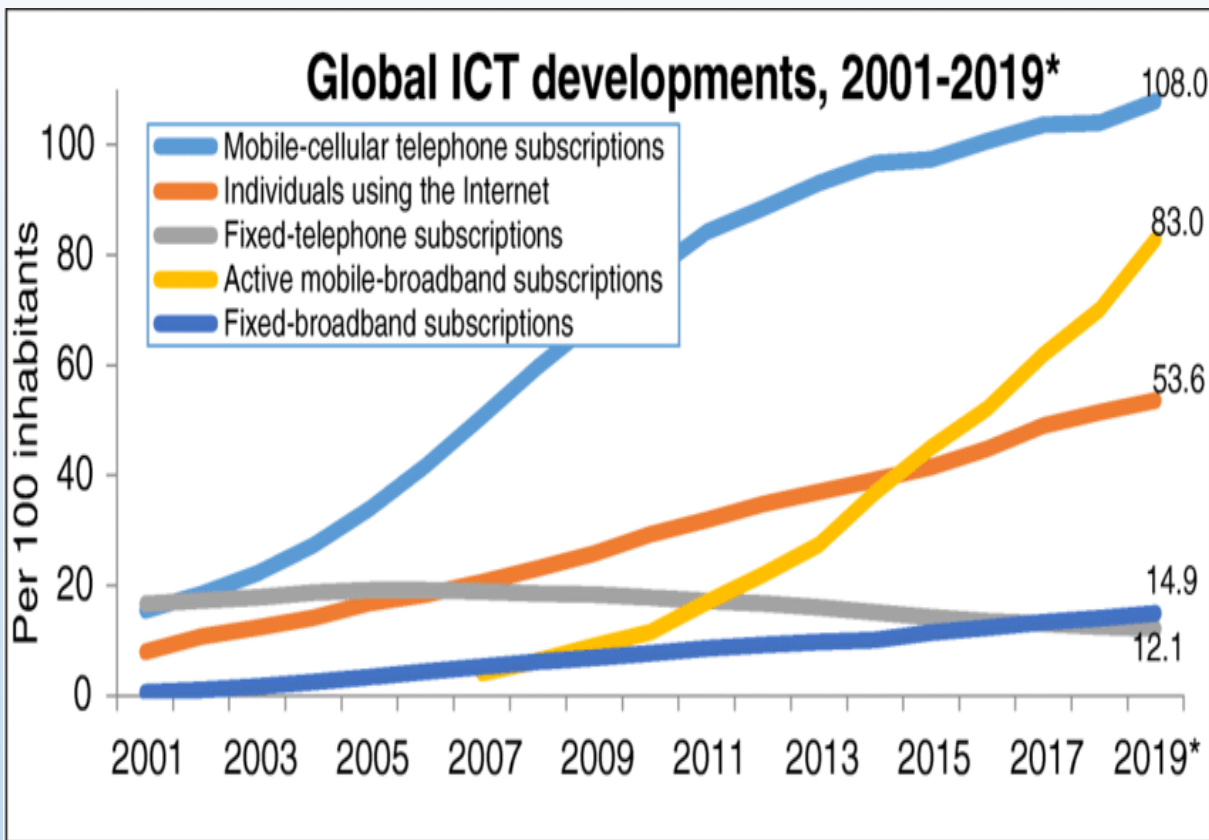
Access

Access is a crucial component in the digital divide. It involves obstacles that exist even if all the other remaining indicators are in place.

These obstacles can broadly be grouped into five categories:

- Geography
- Income
- Ethnicity
- Age and
- Education.

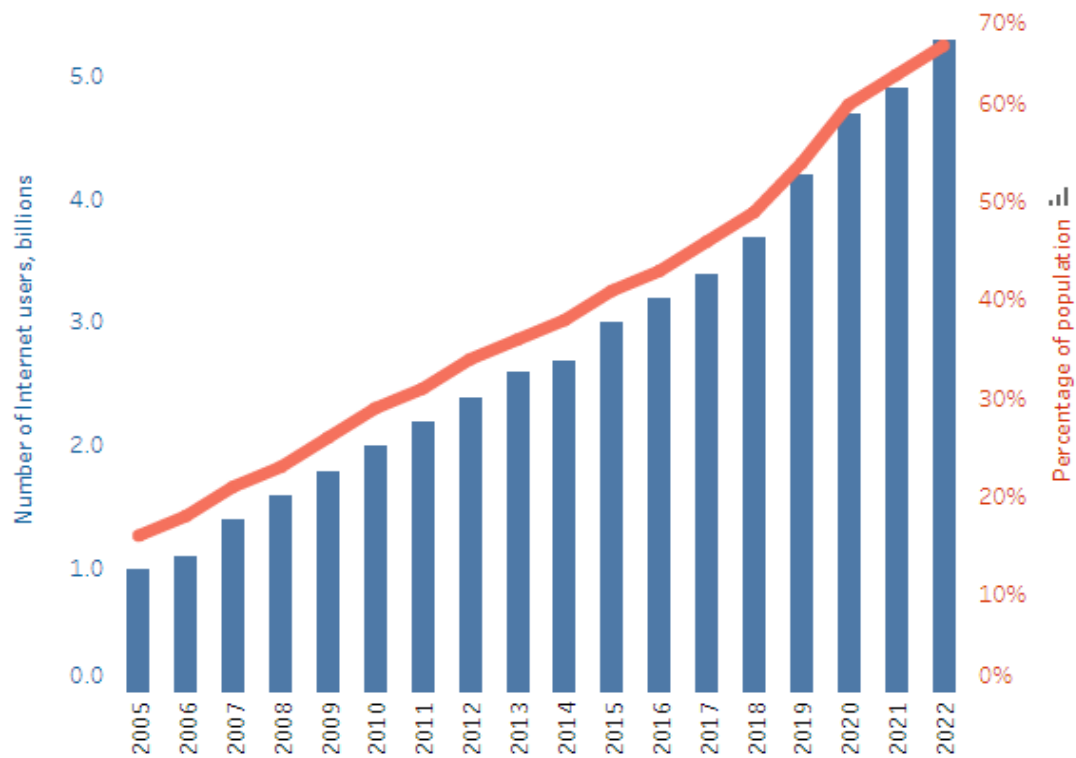




Note: *Estimates Source: ITU World Telecommunication/ICT Indicators database



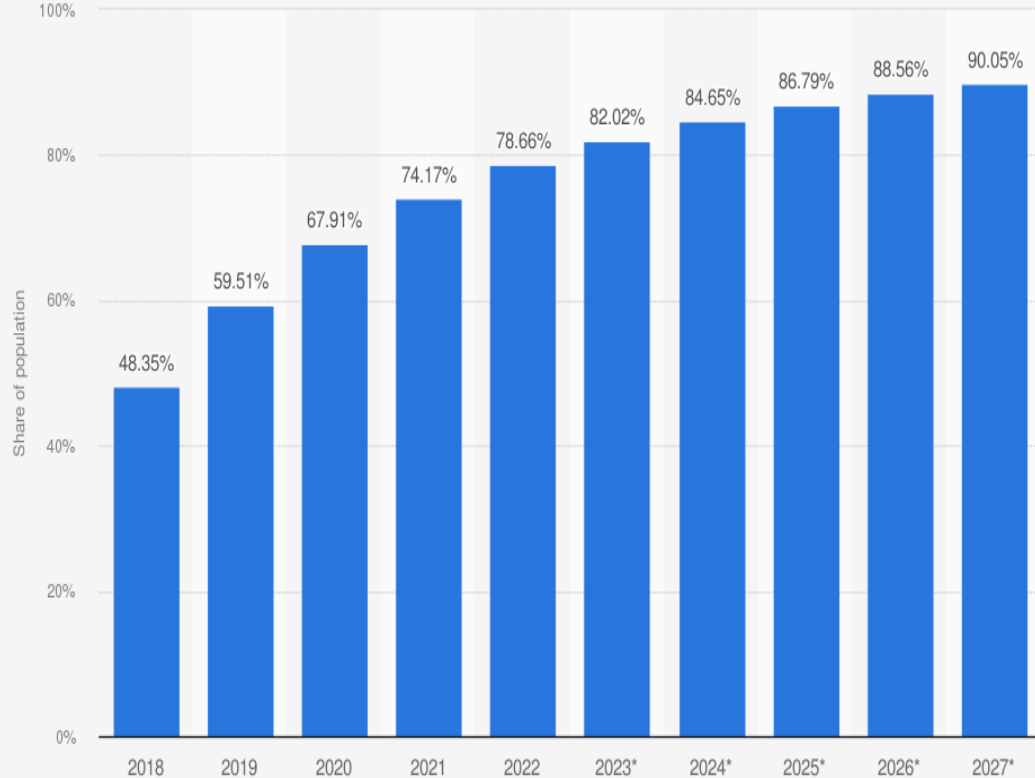
Individuals using the Internet



Source: ITU



Mobile internet user penetration in South Africa from 2018 to 2027

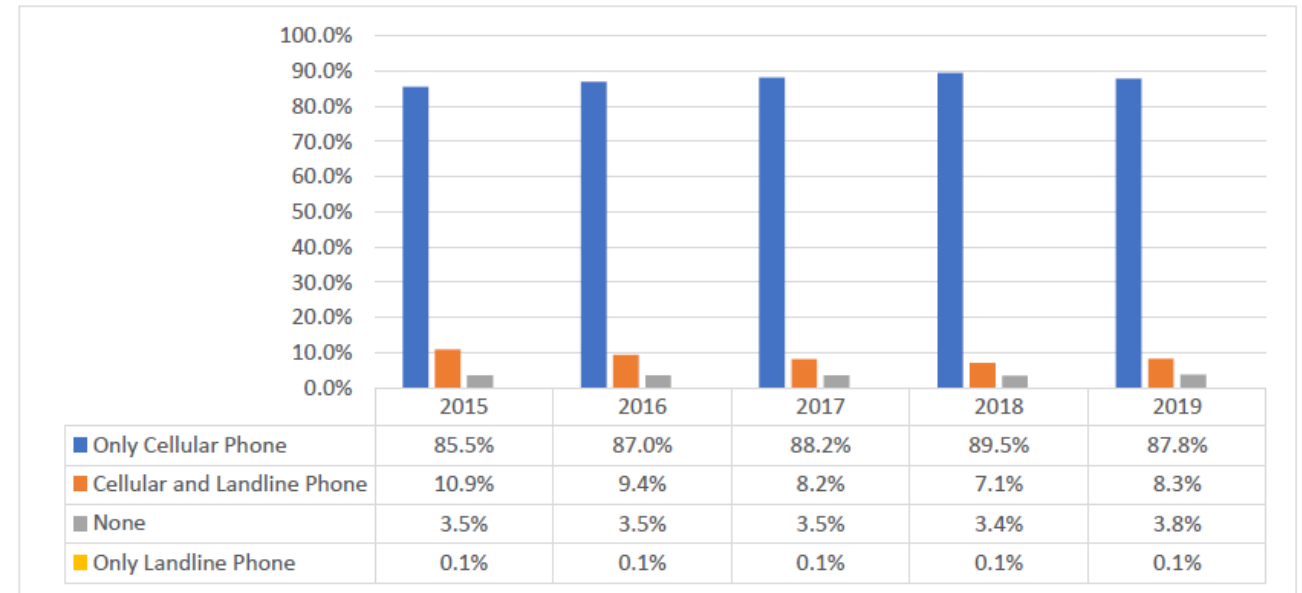


Source
Statista
© Statista 2022

Additional Information:

South Africa; Statista; 2018 to 2022; users who access the internet via mobile broadband connection regardless of the device

Graph 1: Proportion of households who have a functional landline and/or cellular telephone in the Republic of South Africa for 2015 to 2019

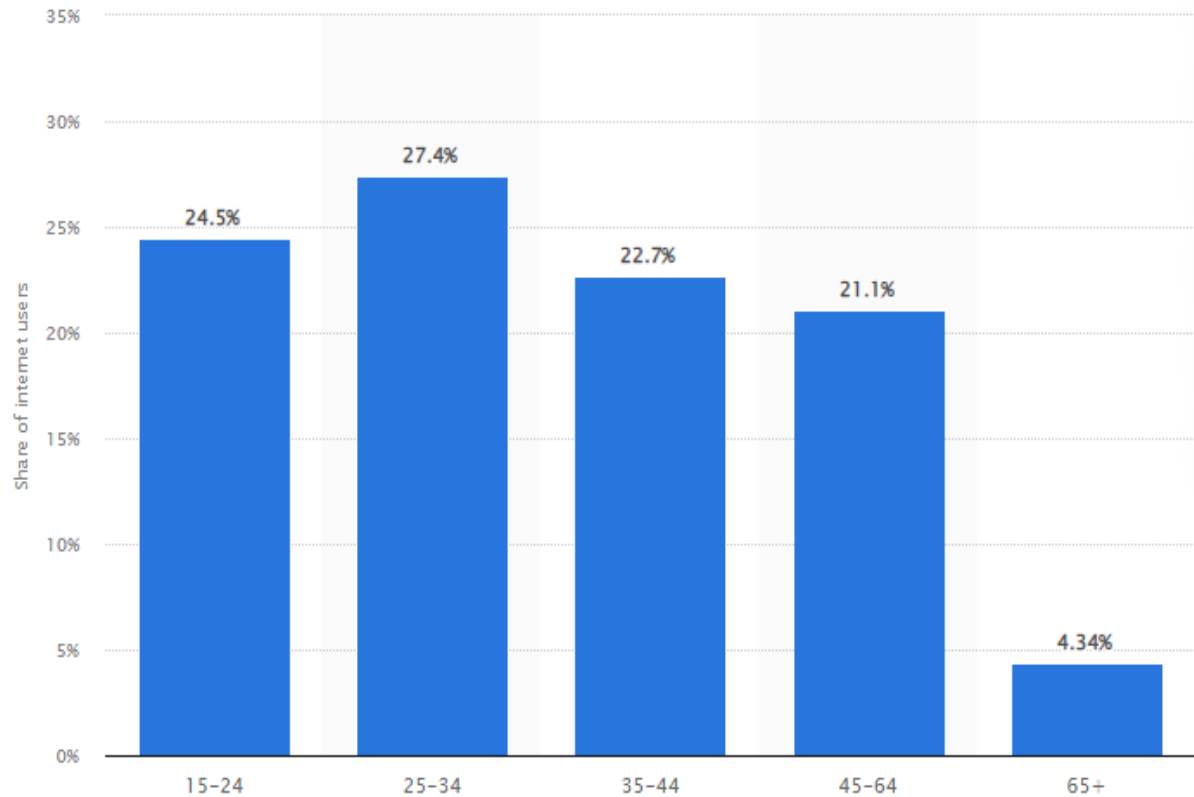


Source: StatsSA GHS, 2015,2016,2017,2018 & 2019

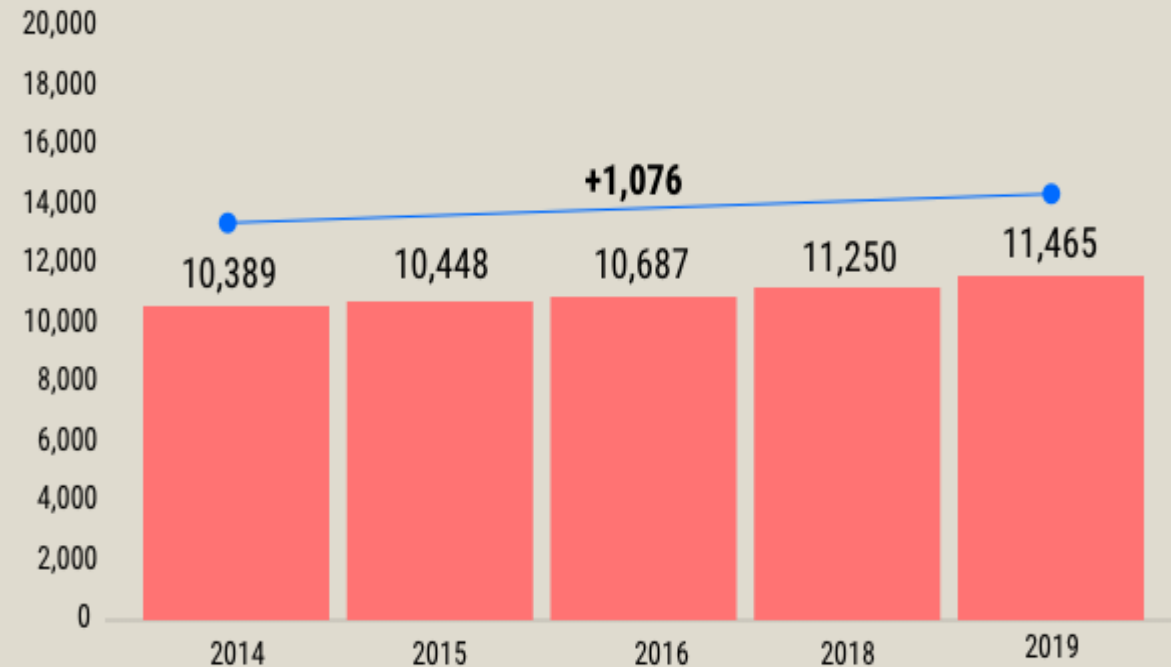
<https://www.statista.com/statistics/972866/south-africa-mobile-internet-penetration/>

The State of the ICT Sector Report in South Africa 2021

Distribution of internet users in South Africa as of 2021, by age group



Internet access in SA schools 2014 - 2019



<https://mediahack.co.za/datastories/schools-internet/>

Technology

The computer-driven technological revolution has brought the countries of the world closer together. In their study of the digital divide, Rodriquez and Wilson observed that all developing countries, including the poorest, are improving their access to the use of ICT.

Investment of Technology is usually in the form of :

- Hardware
- Software



Human ware (Human Capacity)

Human capacity development is a complex multifaceted endeavor consisting of many parts including :

- Creating awareness of the potential for ICT to meet one's needs
- Creating, developing, and strengthening capacity to use information and ICT effectively, using local inputs
- Building capacity to produce and package information so that it adds value to local inputs
- Ensuring ongoing technical capacity development and developing a format for knowledge and information sharing
- Preventing the local capacity from being drained to other, usually developed countries



Infrastructure

- Infrastructure will mean fixed communication structures countries with good fixed communication structures like electricity, telephones, good roads, and airports, ICT development is a lot faster. Lack of such resources hinders the development of ICT.



Enabling Environments

An ICT-enabling environment is an environment in which ICT can thrive.

There are several things that can bring about such an environment:

- **Politics:** A climate of democratic rights and civil liberties conducive to ICT adaptation .Respect for the rule of law and security of property rights. Investment in human capacity.
- **Public policy and Management styles:** Governments must put in place streamlined regulatory policies for the importation and licensing of ICT technologies. Laws must be enacted and enforced uniformly so that nongovernmental organizations (NGOs) and other organizations interested in investing in ICT economic activities do so with ease.



ICT in the Workplace



The automation of the workplace has been the most vigorously pursued concept since the industrial age. Despite the original fear that workplace automation would mean the end to human work, except in a few areas, workplace automation has proceeded hand in hand with increases in employment numbers.

- The Electronic Office
- Office on Wheels and Wings
- The Virtual Workplace

Categories of Telecommuters

There are three categories of telecommuters.

- Consists of workers who use their homes as an adjunct to their conventional office jobs.
- Consists of workers who use their homes as the base for their businesses.
- Consists of those who have full-time jobs with large companies but prefer through their own initiative to work from home.



Effects and Benefits of Telecommuting

- Gender
- Nature of Work
- Labor supply
- Age



Employee Monitoring

- Management is using a wide array of surveillance gadgets and techniques. These include, among others, employees taking polygraph tests if they are suspected of a breach of any kind.
- Electronic monitoring is generally the monitoring of employees using electronic devices like video cameras, computer equipment, audio devices, and many other concealed gadgets. In most cases, it measures the quality and usually the quantity of work and the ability and effectiveness of the worker. In other cases, it also measures



Consequences of Electronic Monitoring

Electronic monitoring also causes the following problems:

- Reduced task variety
- Lack of individual initiatives
- Reduced or no peer social support
- Lack of self – esteem
- Lack of interest in the job
- Lack of trust among workers , between workers and supervisors and between supervisors and management



Workplace, Employee, Health, and Productivity

- Ergonomics: Ergonomics is an applied science concerned with designing human–machine interactions that offer and maintain a safe, comfortable, healthy, and habitable work environment.
- Repetitive Strain Injuries
- Stress

